Inventory Management

Types of Inventories

- ☑ Raw material
  - ☑ Purchased but yet not processed
- ☑ Work-in-process
  - ☑ Undergone some change but not completed
- ☑ Maintenance/repair/operating (MRO)
  - ☑ Necessary to keep machinery and processes productive
- ☑ Finished goods
  - ☑ Completed product awaiting shipment

Functions of Inventory

- ☑ To meet variation in product demand
- ☑ To allow flexibility in production scheduling
- ☑ To protect against stock-outs
- ☑ To take advantage of economic purchase order size
- ☑ To help hedge against price increases
- ☑ To provide a safeguard for variation in raw material delivery time

Effective Inventory Management

- ☑ ***A system to keep track of inventory***
- ☑ ***A reliable forecast of demand***
- ☑ ***Knowledge of lead times***
- ☑ ***Reasonable estimates of***
  - ***Holding costs***
  - ***Ordering costs***
  - ***Shortage costs***
- ☑ ***A classification system***

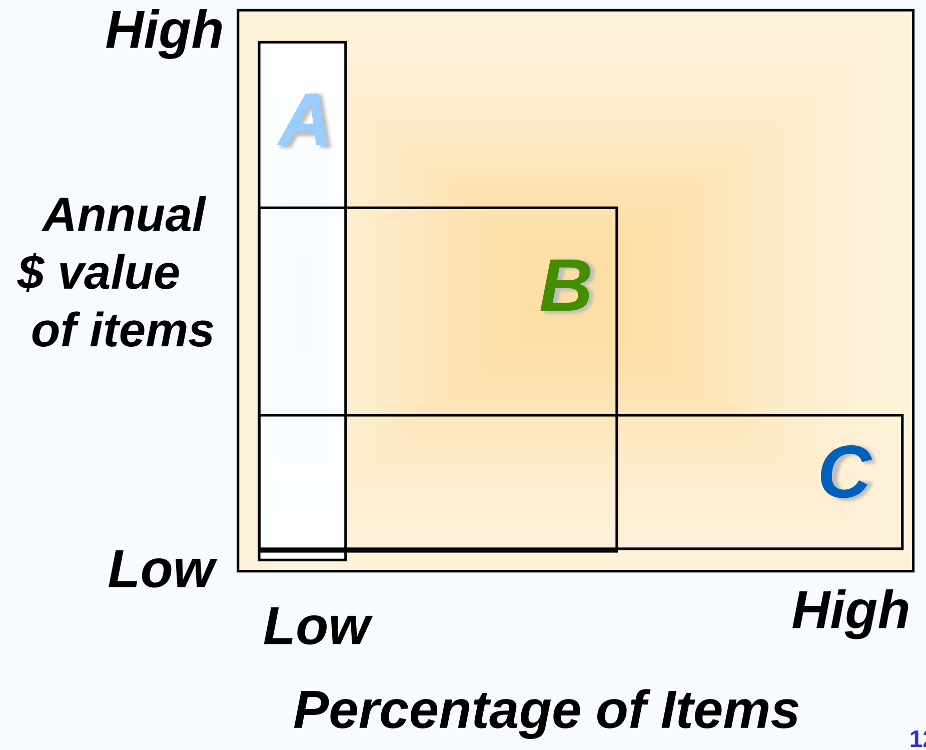
Key Inventory Terms

- ☑ ***Lead time:*** *time interval between ordering and receiving the order*
- ☑ ***Holding costs:*** *the costs of holding or "carrying" inventory for a length of time, usually a year*
- ☑ ***Ordering costs:*** *the costs of placing an order and receiving goods*
- ☑ ***Setup costs:*** *cost to prepare a machine or process for manufacturing an order*
- ☑ ***Shortage costs:*** *costs when demand exceeds supply*

ABC Classification

Classifying inventory according to some measure of importance and allocating control efforts accordingly.

- ***A** - very important*
- ***B** - mod. important*
- ***C** - least important*



ABC Classification

Item Number	Annual Dollar Usage	Percentage of Total Value
22	$ 95,000	40.69%
68	75,000	32.13
27	25,000	10.71
03	15,000	6.43
82	13,000	5.57
54	7,500	3.21
36	1,500	0.64
19	800	0.34
23	425	0.18
41	225	0.10
	$233,450	100.0%

Classification	Item Number	Annual Dollar Usage	Percentage of Total
A	22, 68	$170,000	72.9%
B	27, 03, 82	53,000	22.7
C	54, 36, 19, 23, 41	10,450	4.4
		$233,450	100.0%

Inventory Models for Independent Demand

Need to determine when and how much to order

- ☑ Economic order quantity model
- ☑ Economic production quantity model
- ☑ Quantity discount model

Basic EOQ Model Assumptions

1. Only one product is involved
2. Annual demand requirements known
3. Demand is even throughout the year
4. Lead time (time from ordering to receipt) is known and constant
5. Each order is received in a single delivery
6. Quantity discounts are not possible that means price per unit of product is constant.
7. Inventory holding cost is based on average inventory.
8. Ordering or setup costs are constant.

Inventory Usage Over Time

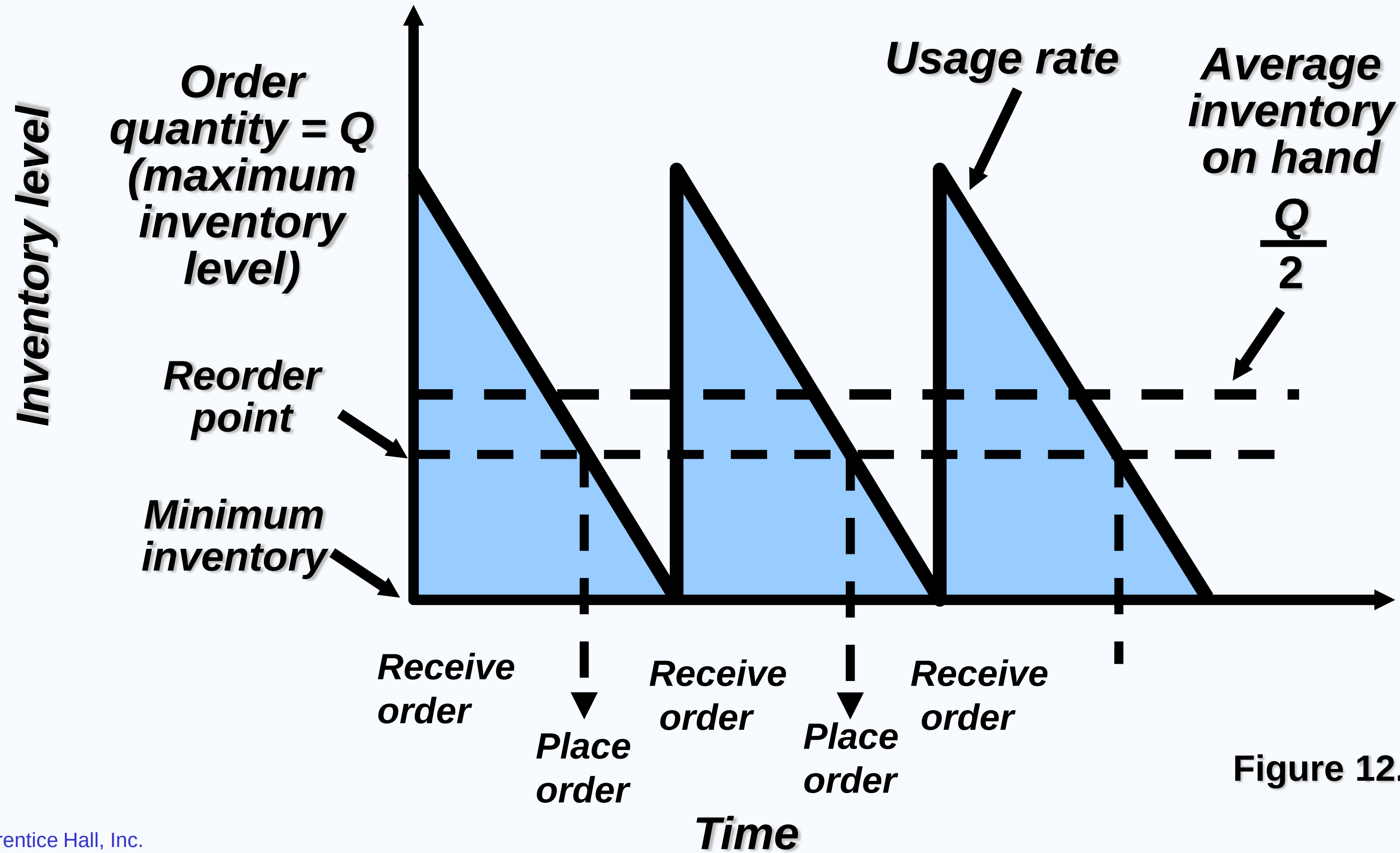


Figure 12.3

Minimizing Costs

Objective is to minimize total costs

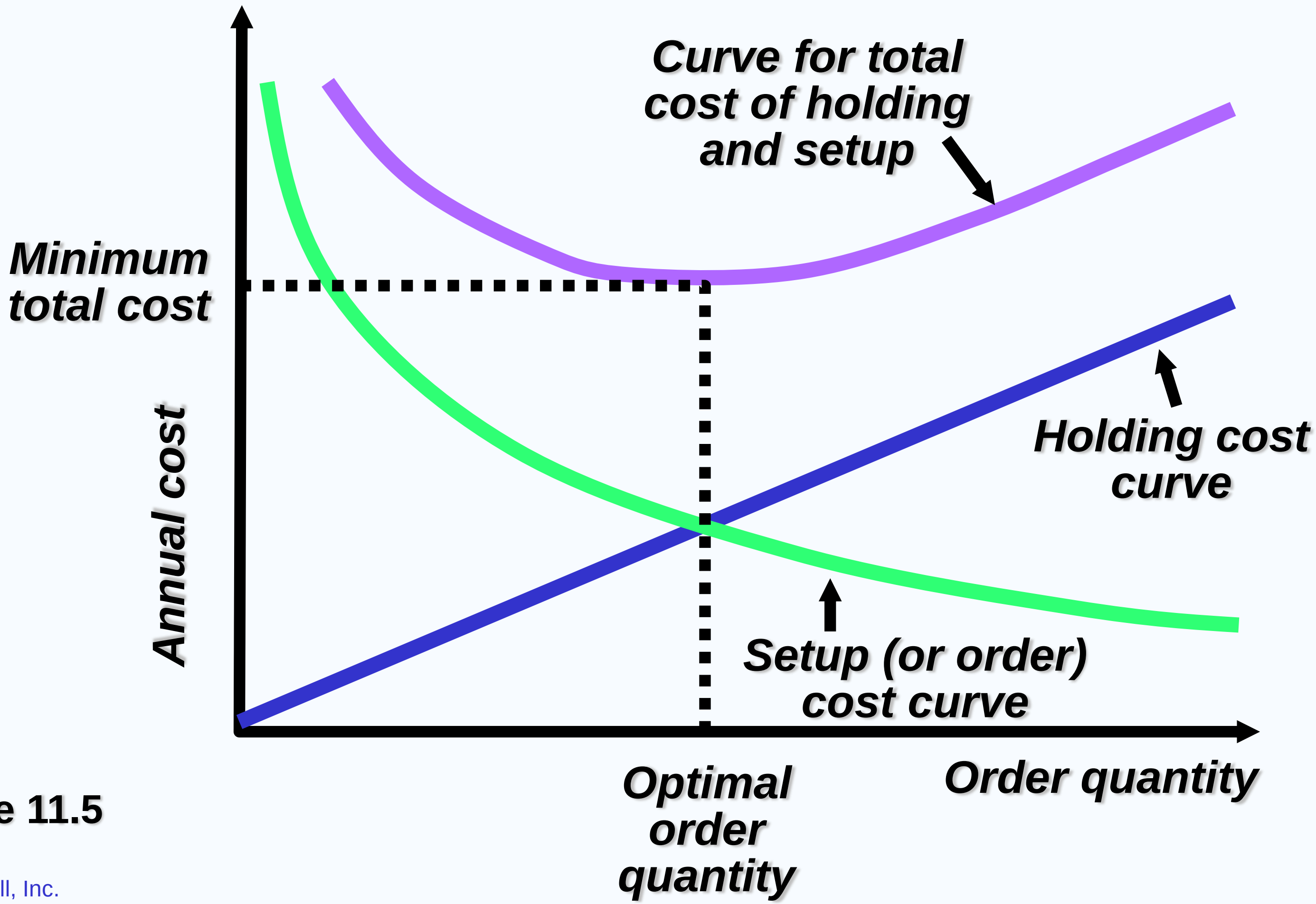


Table 11.5

The EOQ M

Annual setup cost = $\frac{D}{Q}S$

Q = *Number of pieces per order*
*Q** = *Optimal number of pieces per order (EOQ)*
D = *Annual demand in units for the Inventory item*
S = *Setup or ordering cost for each order*
H = *Holding or carrying cost per unit per year*

Annual setup cost = (*Number of orders placed per year*) x (*Setup or order cost per order*)

$$= \left(\frac{\text{Annual demand}}{\text{Number of units in each order}}\right)\left(\begin{array}{c}\text{Setup or order}\\ \text{cost per order}\end{array}\right)$$

$$= \left(\frac{D}{Q}\right)(S)$$

The EOQ Mo

$$\textit{Annual setup cost} = \frac{D}{Q}S$$

$$\textit{Annual holding cost} = \frac{Q}{2}H$$

***Q* = Number of pieces per order**
***Q** = Optimal number of pieces per order (EOQ)**
***D* = Annual demand in units for the Inventory item**
***S* = Setup or ordering cost for each order**
***H* = Holding or carrying cost per unit per year**

$$\textit{Annual holding cost} = (\textit{Average inventory level}) \times (\textit{Holding cost per unit per year})$$

$$= \left(\frac{\textit{Order quantity}}{2}\right)(\textit{Holding cost per unit per year})$$

$$= \left(\frac{Q}{2}\right)(H)$$

The EOQ Mo

$$\text{Annual setup cost} = \frac{D}{Q}S$$

$$\text{Annual holding cost} = \frac{Q}{2}H$$

Q = Number of pieces per order
Q^* = Optimal number of pieces per order (EOQ)
D = Annual demand in units for the Inventory item
S = Setup or ordering cost for each order
H = Holding or carrying cost per unit per year

Optimal order quantity is found when annual setup cost equals annual holding cost

$$\frac{D}{Q}S = \frac{Q}{2}H$$

Solving for Q^*

$$2DS = Q^2H$$

$$Q^2 = 2DS/H$$

$$Q^* = \sqrt{2DS/H}$$

An EOQ Example

Determine optimal number of needles to order

D* = 1,000 *units

S* = $10 *per order

H* = $.50 *per unit per year

$$Q^* = \sqrt{\frac{2DS}{H}}$$

$$Q^* = \sqrt{\frac{2(1{,}000)(10)}{0.50}} = \sqrt{40{,}000} = 200 \text{ units}$$

An EOQ Example

Determine optimal number of needles to order

D = 1,000 *units* Q^* = 200 *units*

S = $10 *per order*

H = $.50 *per unit per year*

$$\text{Expected number of orders} = N = \frac{\text{Demand}}{\text{Order quantity}} = \frac{D}{Q^*}$$

$$N = \frac{1{,}000}{200} = 5 \text{ orders per year}$$

An EOQ Example

Determine optimal number of needles to order

D = 1,000 *units*

S = \$10 *per order*

H = \$.50 *per unit per year*

Q^* = 200 *units*

N = 5 *orders per year*

$$\text{Expected time between orders} = T = \frac{\text{Number of working days per year}}{N}$$

$$T = \frac{250}{5} = 50 \text{ days between orders}$$

An EOQ Example

Determine optimal number of needles to order

D = 1,000 *units*

S = \$10 *per order*

H = \$.50 *per unit per year*

Q^* = 200 *units*

N = 5 *orders per year*

T = 50 *days*

Total annual cost = Setup cost + Holding cost

$$TC = \frac{D}{Q}S + \frac{Q}{2}H$$

$$TC = \frac{1{,}000}{200}(\$10) + \frac{200}{2}(\$.50)$$

$$TC = (5)(\$10) + (100)(\$.50) = \$50 + \$50 = \$100$$

Reorder Points

☑ *The reorder point (ROP) tells when to order*

$$ROP = \begin{pmatrix}\text{Demand} \\ \text{per day}\end{pmatrix}\begin{pmatrix}\text{Lead time for a} \\ \text{new order in days}\end{pmatrix}$$

$$= d \times L$$

$$d = \frac{D}{\text{Number of working days in a year}}$$

Reorder Point Example

Demand = 1,000 *per year*
250 *working days per year*
Lead time for orders is 3 *working days*

$$d = \frac{D}{\text{Number of working days in a year}}$$

$$= 1{,}000/250 = 4 \textit{ units}$$

$$\text{ROP} = d \times L$$

$$= 4 \textit{ units per day} \times 3 \textit{ days} = 12 \textit{ units}$$

Quantity discount model Example 1

The maintenance department of a large hospital uses about 816 cases of liquid cleanser annually. Ordering costs are \$12, carrying costs are \$4 per case a year, and the new price schedule indicates that orders of less than 50 cases will cost \$20 per case, 50 to 79 cases will cost \$18 per case, 80 to 99 cases will cost \$17 per case, and larger orders will cost \$16 per case. Determine the optimal order quantity and the total cost.

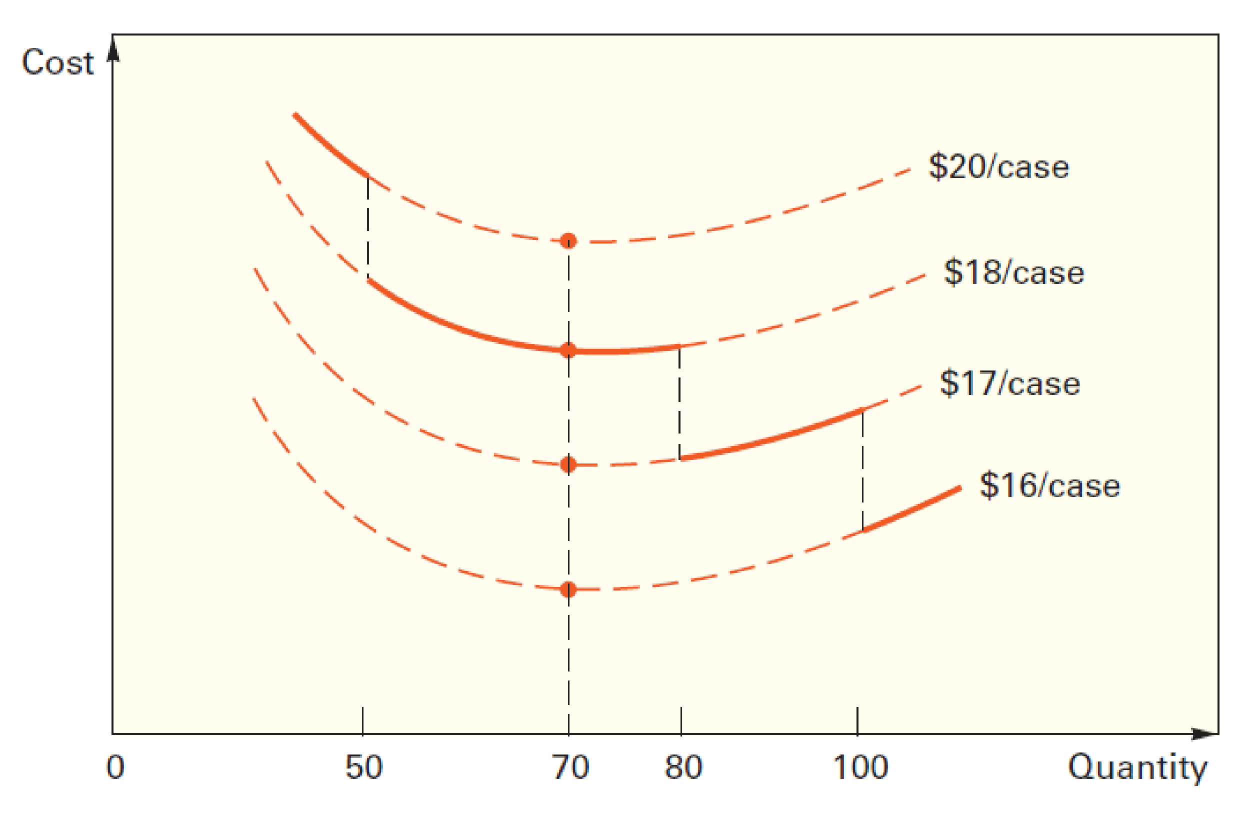
See Figure 13.10:

$D = 816$ cases per year $S = \$12$ $H = \$4$ per case per year

Range	Price
1 to 49	\$20
50 to 79	18
80 to 99	17
100 or more	16

Example 1: Solution

1. Compute the common minimum Q: $= \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2(816)12}{4}} = 69.97 \approx 70$ cases.



Example 1: Solution

2. The 70 cases can be bought at \$18 per case because 70 falls in the range of 50 to 79 cases. The total cost to purchase 816 cases a year, at the rate of 70 cases per order, will be

$$
\begin{aligned}
TC_{70} &= \text{Carrying cost} + \text{Order cost} + \text{Purchase cost} \\
&= (Q/2)H + (D/Q)S + PD \\
&= (70/2)4 + (816/70)12 + 18(816) = \$14,968
\end{aligned}
$$

Because lower cost ranges exist, each must be checked against the minimum cost generated by 70 cases at \$18 each. In order to buy at \$17 per case, at least 80 cases must be purchased.

Example 1: Solution

(Because the TC curve is rising, 80 cases will have the lowest TC for that curve's feasible region.) The total cost at 80 cases will be

$$TC_{80} = (80/2)4 + (816/80)12 + 17(816) = \$14,154$$

To obtain a cost of $16 per case, at least 100 cases per order are required, and the total cost at that price break will be

$$TC_{100} = (100/2)4 + (816/100)12 + 16(816) = \$13,354$$

Therefore, because 100 cases per order yields the lowest total cost, 100 cases is the overall optimal order quantity.

Quantity discount model Example: 2

Cong Manufacturing has gone out to bid for a regulator component. Expected demand is 700 units per month. The item can be purchased either from Vesey Manufacturing or from Vaccaro manufacturing. Their price lists are shown below. Ordering cost is $50, and annual holding cost per unit is $5.

(a) What is economic order quantity (EOQ)?

(b) Which supplier should be used? Why?

(c) What is the optimal order quantity and the total annual cost (ordering, purchasing and holding the component)?

Quantity	Min	Unit price
1-499	1	16.00
500-999	500	15.50
1000+	1000	15.00

Quantity	Min	Unit price
1-399	1	16.10
400-799	400	15.60
800+	800	15.10

Example 2: Solution

Economic Order Quantity (EOQ)

$$EOQ = \sqrt{\frac{2 \times D \times S}{H}}$$

Where:

D is the demand rate (700 units per month or 8400 units per year in this case)

S is the order cost ($50 in this case)

H is the holding cost per unit per year ($5 in this case)

Substituting the given values into the formula, we get:

$$EOQ = \sqrt{\frac{2 \times 8400 \times 50}{5}}$$

$=410$

Example 2: Solution

For any discount, if the order quantity is too low to quantify for the discount, adjust Q* upward to the lowest feasible quantity.

For Vesey:

Quantity	Min	Unit price	Q*	Adjusted Q*
1-499	1	16.00	410	410
500-999	500	15.50	410	500
1000+	1000	15.00	410	1000

For Vaccaro:

Quantity	Min	Unit price	Q*	Adjusted Q*
1-399	1	16.10	410	-
400-799	400	15.60	410	410
800+	800	15.10	410	800

Example 2: Solution

Calculate the total annual cost for each Q*.

Total annual cost=Ordering cost + Holding cost

+ Purchasing cost

$$= \frac{D}{Q}S + \frac{Q}{2}H + CD$$

Example 2: Solution

For Vesey:

Quantity	Min	Unit price	Adjusted Q*	Ordering Cost ($)	Holding Cost ($)	Purchasing Cost ($)	Total ($)
1-499	1	16.00	410	1,025	1,025	134,400	136,449
500-999	500	15.50	500	840	1,250	130,200	132,290
1000+	1000	15.00	1000	420	2,500	126,000	128,920

For Vaccaro:

Quantity	Min	Unit price	Adjusted Q*	Ordering Cost ($)	Holding Cost ($)	Purchasing Cost ($)	Total ($)
1-399	1	16.10	410	-	-	-	-
400-799	400	15.60	410	1,025	1,025	131,040	133,089
800+	800	15.10	800	525	2,000	126,840	129,365

It looks like the thing to do is to order 1,000 at a time from Vesey Manufacturing.

Quantity discount model Example: 3

- Suppliers of Non-Security Examination Materials, occasionally give discounts to the organization in line with its organizational policy in order to reduce a large stock of its materials which may increase its holding cost. In 2012 it offered a quantity discount pricing schedule (as shown in table below)

Quantity ('000)	Price (Rs. per unit)
1-49	1400
50-89	1100
90-349	900
350 and above	890

The annual carrying cost for the organization for such materials is 190, the ordering cost is 2,500, and annual demand for this particular model is estimated to be 200 units. The management wants to determine if it should take advantage of this discount or order the basic EOQ order size.
We first determine the optimal order size and total cost with the basic EOQ model.
S = 2,500, H = 190 per material, D = 200 materials per year.

Answer: Q*=73,
TC (73)=RS. 233,784, TC (90) = Rs. 194, 105, TC (350)= 212,678.57.